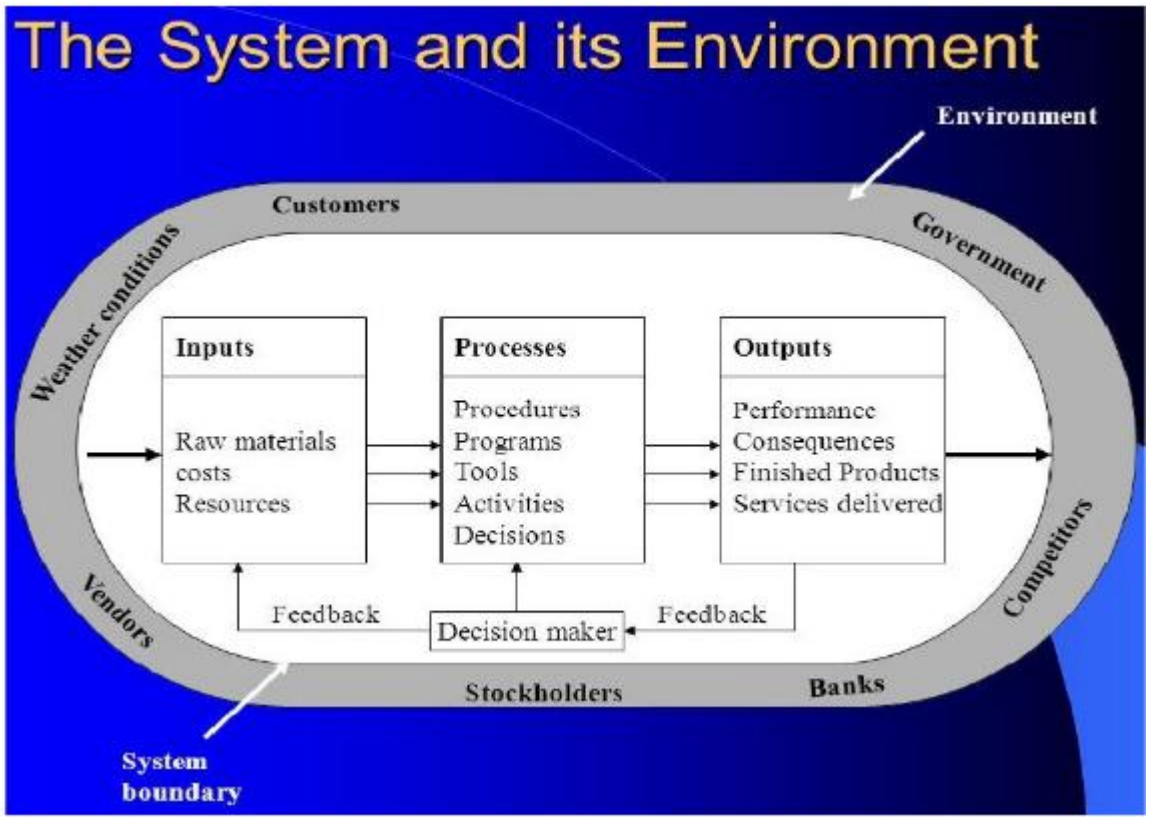
INTRODUCTION

1.1 Systems

The term system is used in many technology-related concepts including Decision Support System and Transaction Processing System….both are Information Systems.

A ***system*** is an interrelated set of components including people, activities, technology and procedures that are designed or intended to achieve a predefined purpose. A system receives **input (s)** from its environment and the various subsystems or components of the system **interact** to produce **outputs.** System components are surrounded by an imaginary boundary that separates a specific system from its environment. A system designer identifies both inputs from the environment as well as the outputs from the system.

System Illustration



A system has a feedback mechanism to provide a means of controlling the operation of the system. Feedback is an input from a system that later re-enters the system as an input.

Example

Let's examine a simple conceptual specification of a system. The initial input into the system is a bank customer requesting a loan. The customer makes a request to a bank officer. The bank officer collects information from the customer and enters that information into a computerized form. A loan approval model is built into a computerized decision aid. Some people identify the computerized model as the actual decision support system. The banker

uses the result from the computerized loan approval model to finalize the decision to approve or deny the loan. In some cases the loan information will need to be shared with a loan committee possibly using a group support system.

The actual decision is then communicated to the customer either face-to-face or by a formal letter that may be generated by a computerized decision aid. Feedback comes from the customer.

This decision process and the overall conceptual system may include various Decision Support Systems. The bank's Transaction Processing System would be updated when the loan was made and the funds disbursed. The loan is the primary transaction. Making the loan is the decision process. DSS can support making loans or a DSS can can help analyse lending activity at the bank or predict activity and interest rates.

In DSS, the primary focus is often on the computerized components of the system. This is a narrow perspective for defining the components of a system; it is often helpful to define the DSS boundary to include a broader decision process that may involve people performing non-computerized tasks as well as more routine data gathering tasks. The users of the computerized tools are also part of the broader system. Finally, note that the actual communication or transmission of decisions may not occur using computerized systems.

This step in a decision process needs to be considered in the design of the DSS and it should be included within the boundary of the system.

1.2 Information systems and its concept

Decision: is a conclusion or resolution reached after consideration. A choice made between alternative courses of action in a situation of uncertainty.

Knowledge – awareness or familiarity gained by experience of a fact or situation. Knowledge comprises things that are held to be true in a given context and that drive us to action if there were no impediments.

Information – the processed data is an increment in knowledge. It contributes to the general framework of concepts and facts that we know. Information relies on the context and the recipient's general knowledge for its significance.

Data – the raw facts, the material for obtaining information. Information systems use data stored in computer database to provide needed information.

Database – an organized collection of logically interrelated data reflecting a major aspect of a firm's activities.

1.3 Information and its role in Business

- Information systems capture data from the organization (internal data) and its environment (external data).
- They store the database items over an extensive period.

- ❖ When specific information is needed, the appropriate data items are manipulated as necessary and the user receives the resulting information.
- ❖ Depending on the type of information system, the information output may take the form a query response, decision outcome, expert-system advice, transaction document, or a report.

Formal information systems rely on procedures (established and accepted by organizational practice) for collecting, storing, manipulating, and accessing data in order to obtain information. Formal systems do not have to be computerized, but today they usually are. Informal information systems also exist within an organization. ***Quality information*** needs to possess several attributes/properties. Notably, it has to be:

i) **Timely:** available when needed and not out-dated when made available.
ii) **Complete:** includes all the user needs to know about the situation where the information will be used.
iii) **Concise:** Does not include elements unneeded by the user
iv) **Relevant:** Has direct bearing on the situation
v) **Form:** the level of detail, tabular versus graphic display, and quantitative versus qualitative form are selected in accordance with the situation internal and external information.

Most of the data captured by information systems relates to the operations of the organization itself, serving to produce internal information. But in an increasingly competitive marketplace, a firm needs more and more external information.

A firm can succeed only by adapting itself to the demands of its external environment. The environment is represented by a number of groups that affect the company's ability to achieve its objectives or that is affected by it. Such groups are called the stakeholders of a frim, which includes both internal and external stakeholders.

A ***system*** is a set of components (subsystems) that operate together to achieve certain objectives. The objectives of a system are realized in its outputs. In particular, the objective of an information system is to provide the appropriate outputs to the members of an organization.

1.4 Components of Information Systems

Information system consists of the following components:



a) **Hardware** – multiple computer systems: microcomputers, minicomputers, mainframes, together with their peripherals. Computer system components are CPU, memory, input/output devices and storage devices.

b) **Software** – the system and application software. System software manages the computer system resources and simplifies programming while application software directly assists end users in doing their work.

   The DSS software system contains the software tools that are used for data analysis. It may contain various OLAP tools, data-mining tools, or a collection of mathematical and analytical models that easily can be made accessible to the DSS user. A model is an abstract representation that illustrates the components or relationships of a phenomenon. A model can be a physical model (such as a model airplane), a mathematical model (such as an equation), or a verbal model (such as a description of a procedure for writing an order).

c) **Databases -** The DSS database may be a small database residing on a PC or a massive data warehouse. The DSS database is a collection of current or historical data from a number of applications or groups. It may be a small database residing on a PC that contains a subset of corporate data that has been downloaded and possibly combined with external data. Alternatively, the DSS database may be a massive data warehouse that is continuously updated by major corporate TPS (including enterprise systems and data generated by Web site transactions). The data in DSS databases are generally extracts or copies of production databases so that using the DSS does not interfere with critical operational systems.

d) **Human resources** (People) – professional information systems personnel include development and maintenance managers, system analysts, programmers, and operators, often with highly specialized skills. End users are the people who use information systems or their information outputs, that is, the majority of people in today's organizations.

e) **Procedures** – the policies and methods to be followed in using, operating and maintaining an information system. Specifications for the use, operation and

maintenance of information systems, collected in help facilities, user manuals, operator manuals, and similar documents, frequently delivered in an electronic form.

f) **DSS User Interface** - The DSS user interface permits easy interaction between users of the system and the DSS software tools. A graphic, easy-to-use, flexible user interface supports the dialogue between the user and the DSS. The DSS users can be managers or employees with no patience for learning a complex tool, so the interface must be relatively intuitive.

   Many DSS today are being built with Web-based interfaces to take advantage of the Web's ease of use, interactivity, and capabilities for personalization and customization. Building successful DSS requires a high level of user participation to make sure the system provides the information managers need.

1.5 Types of Information Systems

Organizations employ several types of information systems. These include:

1. Transaction Processing Systems (TPS)
2. Management Information Systems (MIS)
3. Decision Support Systems (DSS)
4. Executive Information Systems (EIS)
5. Office Information Systems (OIS)
6. Professional Support Systems
7. Expert Systems (ES)

1. **Transaction Processing Systems (TPS)**

   TPS today generally work in on-line mode by immediately processing a firm's business transactions. A transaction is an elementary activity conducted during business operations. TPS may work either in batch mode (processing accumulated transactions at a single time later on) or in online mode (processing incoming transactions immediately.

2. **Management Information Systems (MIS)**

   The objective of MIS is to provide routine information to managers. Managers receive performance reports within their specific areas of responsibility. Generally there reports provide internal information rather than spanning corporate boundaries. They report on the past and the present, rather than projecting the future. In order to prevent information overloads, managers may resort to using demand or exception reports. Demand reports are requested when needed while exception reports are produced only when pre-established out-of-bounds conditions occur and contain only the information regarding these conditions.

3. **Decision Support Systems (DSS)**

   DSS directly support a decision making session. These systems facilitate a dialog between the user, who is considering alternative problem solutions, and the system that provides built-in models and access to databases. The DSS databases are often extracts from the general databases of the enterprise or from external databases.

4. **Executive Information Systems (EIS)**

EIS support top managers with conveniently displayed summarized information, customized for them. They make a variety of internal and external information readily available in a highly summarized and convenient form. EIS are used to:

- Monitor the performance of the organization
- Assess the business environment
- Develop the strategic directions for the company's future

5. **Office Information Systems (OIS)**
   The main objective of OIS is to facilitate communication between the members of an organization and between the organization and its environment. They include Mailing system, faxing system, Record management system etc. OIS are used to:
   - Help manage documents represented in an electronic format
   - Handle messages such as electronic email, facsimile, and voice mail
   - Facilitate teleconferencing and electronic meetings
   - Facilitate the use of the Internet for communication and access to information.
   - Facilitate the use of task-oriented teams through the use of groupware
6. **Professional Support Systems**
   Professional support systems help in tasks specific to various professions. As both organizational and individual experience with information systems grow, more and more specialized categories of professional support systems emerge. Example, QuickBooks for Accountants.
7. **Expert Systems (ES)**
   ES are system that employs knowledge about its application domain and uses an inference (reason) procedure to solve problems that would otherwise require human competence or expertise. The essential component of the knowledge base is heuristics – informal, judgmental elements of knowledge within the expert system's domain, such as oil exploration or stock valuation. The knowledge base is developed by working with domain specialists. It is further enhanced as the system is used.
   Example: An Expert system in Oncology

1.6 Supporting Managers with Information Systems

A variety of information systems support managers as they play their interpersonal, informational, and decisional roles. The three management-oriented types of systems (MIS, DSS and EIS) provide different kinds of support to the ***three levels of management***:

a) Strategic
b) Tactical
c) Operational

The fundamental functions of management include:

i) **Planning:** establishing goals and selecting the actions needed to achieve them over a specific period of time.

ii) **Controlling:** measuring performance against the planned objectives and initiating corrective action, if needed.
iii) **Leadership:** including the people in the organization to contribute to its goals.
iv) **Organizing:** establishing and staffing an organizational structure for performing business activities

1.7 Examples of DSS Applications

DSS vary in many ways. They differ in terms of who uses a specific system, that is some DSS are used by actual decision makers and some are used by intermediaries like marketing analysts or financial analysts. Some DSS focus on data, some on models and some on communications. DSS also differ in scope, some DSS focus are intended for one "primary" user and used "stand-alone" for analysis and others are intended for many users in an organization.

Major airlines have DSS used by analysts for many tasks including pricing and route selection. Many companies have DSS that aid in corporate planning and forecasting. Specialists often use these DSS that focus on financial and simulation models. Investment evaluation and support systems are increasingly common. FritoLay has a DSS that aids in pricing, advertising, and promotion. Frito-Lay is an American subsidiary of PepsiCo that manufactures, markets, and sells corn chips, potato chips, and other snack foods.

Route salesmen use hand-held computers to support decision making activities. Many manufacturing companies use Manufacturing Resources Planning (MRP) software. This specific operational level DSS supports master production scheduling, purchasing, and materials requirements planning. More recent MRP systems support "what-if" analysis and simulation capabilities.

Monsanto, FedEx and most transportation companies use DSS for scheduling trucks, airplanes and ships. The Coast Guard uses a DSS for procurement decisions. Companies like Wal-Mart have large data warehouses and use data mining software. Business Intelligence and Knowledge Management Systems are increasingly common. On the WWW, one can find DSS that help track and manage stock portfolios, choose stocks, plan trips, and suggest gifts. DSS support distributed decision activities using groupware and a corporate intranet